

Article

Exploring the Role of News Outlets in the Rise of a Conspiracy Theory: Hydroxychloroquine in the Early Days of COVID-19

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Abstract: Improper use of hydroxychloroquine to treat COVID-19 has been linked to 17,000 preventable deaths. This content analysis study investigates the emergence of this conspiracy theory, the role of the news media in perpetuating and disseminating it, and whether coverage differed by outlet political alignment. We searched Nexis for relevant media from 17–31 March 2020. A total of 128 media pieces were coded qualitatively and thematically analysed. The news media amplified the voices of right-wing political elites and used a variety of manipulative tactics in reporting on hydroxychloroquine. Powerful ingroup/outgroup mechanisms polarised the American public and created a schism between Trump supporters and the public health apparatus that reflected the political asymmetry in reporting on hydroxychloroquine. The widespread use of optimistic framings and anecdotal evidence contributed to public misunderstandings of the evidence. Therefore, strategic and interventionist public health efforts are required to combat misinformation. This study informs discussions of how politicised media coverage catalyses conspiracism.

Keywords: public health; health policy; content analysis; media analysis; misinformation; conspiracism; COVID-19



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1. Introduction

Early in the COVID-19 pandemic, in March 2020, fear and anxiety among the public were high [1]. There was no cure or established medical treatment for COVID-19, only care to help with the symptoms. Death rates were increasing [2], and much of the world was in lockdown. Desire for some kind of medical intervention against COVID-19 was very powerful, and hope was something that many clung to out of desperation and fear. It was in this environment that Elon Musk, on 17 March 2020, tweeted about a recent preliminary medical study showing early, anecdotal optimism around a relatively old medical treatment for Malaria, Lupus, and Arthritis known as hydroxychloroquine. Two days later, on 19 March 2020, President Donald Trump tweeted about hydroxychloroquine as well. These instances of a politically conservative elite celebrity tweeting about hydroxychloroquine closely align with dramatic rises in Google search trends around hydroxychloroquine, which highlights the extent of public awareness increase in such a short time [3]. Following this rapid ascent to popularity, hydroxychloroquine became a topic of public discussion and debate. Through this public debate and rapidly evolving and shifting views on hydroxychloroquine from varying social groups, conspiracism and misinformation quickly followed. When following studies showed the ineffectiveness of hydroxychloroquine in treating COVID-19 and the public health apparatus began attempting to rebuff the conspiracism and turn the public away from an ‘alternative’ medical treatment that was not going to help them deal with COVID-19, it was too little, too late. In the end, improper usage of hydroxychloroquine as a treatment for COVID-19 has been estimated to have contributed to nearly 17,000 deaths [4], occurring mainly from diversion from more effective

treatments but with a substantial number of cardiac and other side effects contributing to mortality rates in treatment of COVID-19.

To begin discussing hydroxychloroquine as a treatment for COVID-19, it is essential to first outline the scientific consensus thereon. Although initial studies showed a modicum of promise [5] garnering high public attention, the overwhelming scientific consensus following intensive further study is that hydroxychloroquine has no significant effect as a treatment for COVID-19 and is not recommended for such use [6–8]. Such consensus is the basis for treating this mass public interest in hydroxychloroquine as a treatment for COVID-19 as misinformation that coalesced into a conspiracy theory, as outlined by the narrative central to this study. COVID-19 misinformation surrounding hydroxychloroquine has garnered much study since 2020. The first relevant study to discuss is Sattui et al. [9] who describe what happened as a ‘pendulum’. Early research into hydroxychloroquine as a treatment for COVID-19 involved rapidity in testing and heavy use of preprints. Overinterpretation and amplification of preliminary results outside of the scientific community—followed by attempted corrective measures by the public health apparatuses—created a pendulum effect in which public discourses were initially very supportive of hydroxychloroquine but then swung back far enough to create a backfire effect. This backfire effect is theorised to occur when attempts to dissuade someone of strongly held beliefs misalign with their worldview so much that it reinforces their pre-existing beliefs and degrades trust in legitimate information [10]. Baker & Maddox [11] extend this idea by tracing how the politicisation and sudden rise to public awareness of hydroxychloroquine contributed to a backfire effect in the reactive debunking attempts of the public health apparatus. However, a literary gap exists around the role of the news media. This is particularly relevant given the recent scholarship highlighting the prominent role of cable news and television media in the dissemination of misinformation [12–15].

Social identity theory helps define complex psychological mechanisms of relevance to this study, in particular ingroup/outgroup dynamics. These dynamics are well established in the psychological literature, in which members of social groups present more positive behaviours towards other ‘ingroup’ members of their social group, and contrastingly present negatively towards ‘outgroup’ people who are *not* in their social group [16]. These behaviours can go beyond ‘negative’, however, and have been linked to dehumanisation and political extremism [17]. These effects are particularly significant around radical political groups [18–20] and conspiracism [21,22]. As relates to COVID-19, the literature on ingroup dynamics has mainly focused on national ingroups and outgroup prejudices [23,24], with little yet examining the role of ingroup dynamics around the misinformation and conspiracism so prominent in public perceptions of COVID-19. When engaging with the political elements of COVID-19, however, the literature agrees upon the central role of social identity theory and ingroup/outgroup mechanisms in contemporary political radicalisation, the contemporary American right wing, and even directly ties it to understandings of political support for Donald Trump [25,26].

This study will investigate the role of the U.S. news media in this rise to public awareness and conspiracism around hydroxychloroquine in March 2020. It will explore whether differences in political ideology alignment among media outlets impacted reporting, usage of language, and message framing on this topic. The knowledge gained in this study can inform public health decisions in future pandemics and contribute to the growing body of literature discussing misinformation and interventions thereagainst. Further, this study sheds light on the role of political ideological alignment in media sources as relates to misinformation, through the case study of hydroxychloroquine and COVID-19.

The central research question for this study is: **How did news media reporting on hydroxychloroquine in the period immediately following its rise to popularity (17–19 March 2020) contribute to the subsequent rise of conspiracism and the spread of this essential piece of misinformation?**

Sub questions include:

1. Did coverage of hydroxychloroquine in this period differ substantially by political ideology (proxied via political ideological ranking of the news outlet)?
2. How did the U.S. public health apparatus attempt to combat this misinformation in this period?

2. Methods

This study followed an adjusted version of the PROSPERO protocol for pre-registration, excluding the elements that were not applicable to this study as content analysis instead of systematic review. The study submitted a pre-registration with Zenodo on 11 March 2024, with the registration number 10804505 and the record title “Tracing the role of news outlets in the rise of a conspiracy theory: Hydroxychloroquine in the early days of COVID-19”, available at the Zenodo website.

This study utilises a content analysis methodology, which is similar to, yet distinct from a systematic review. While both approaches share a goal of synthesising existing information, content analysis focuses on categorising and interpreting qualitative data from media sources, rather than synthesising evidence from existing scientific studies. Content analysis allows for the examination of media tone, framing, recurring themes, and manipulative tactics in ways that systematic reviews of scientific studies cannot. Furthermore, content analysis is intended to achieve a complete perspective of media content available within the boundaries of the study. In this study, these boundaries are mainly determined temporally, wherein the comparatively narrow temporal window under study allows for a completionist review of all media content in that period. This study is intended to bolster and supplement systematic reviews of COVID-19 misinformation, focusing on media content and the framing of messages intended for the general public.

2.1. Ethics

All data used in this study were obtained through the publicly accessible Nexis database. As the data are publicly available and do not involve human participants or private information, no ethical approval was required for this research.

2.2. Study Design

A systematic search of online news media was conducted, followed by deductive thematic and content analysis of eligible news articles.

2.3. Search Strategy

This study included a systematic, replicable search of the Nexis database to identify English language news content published between 17 March 2020 and 31 March 2020, performed following the pre-registered protocol available at Zenodo. Nexis is a 40-year archive of more than 40,000 curated news and business sources (both print and transcripts of television broadcasts). Nexis was chosen for ease of institutional access, availability of major media outlets, its extensive catalogue of media content, and its advanced searching systems. Further data collection monitoring systems were not utilised due to the comprehensiveness of data available on Nexis alongside concerns by the author of keeping the study manageable and restricting the data to a functionally workable number of entries. No secondary searches were performed due to the perceived comprehensiveness offered by the first search, alongside practical concerns around keeping the amount of media data manageable for in-depth analysis in a reasonable timeframe. Furthermore, no new data can be generated within the bounds of this study, as the highly bounded temporal window is concrete and in the past.

Search strategy followed the protocol using pre-determined search terms, with results imported into Excel sheets for ease of deselection. Duplicates were removed and then an initial title-based screening was performed. Screening then followed based on headline and then full-text review. Duplicate screening was performed by a team member (K.G.) on ~15%

of studies through all screening stages, with any disagreement resolved via discussion. Inter-rater agreement was found to be very high (~94%).

The following search terms are performed as Boolean searching within the Nexis NEWS database: publication (daily caller OR spectator OR newsbusters OR Fox news OR national review OR realclearpolitics OR FORBES OR National interest OR associated press OR Politico OR CNN OR new york times OR newsweek OR guardian OR new yorker OR salon OR quartz OR the nation OR vanity fair OR Prospect) and hydroxychloroquine.

Eligible News Outlets

To make the search targeted and manageable, a small number of outlets were chosen from the Nexis database. These outlets were pre-identified as influential and widely consumed by their respective audiences, as well as their inclusion in both the Nexis database and the Ideology of Online News Media scale. By including outlets across the political spectrum, the study was able to then capture potential differences in the presentation of information on hydroxychloroquine to the general public audience. The Ideology of Online News Media scale [27] was used to identify a range of news outlets that sit on distinct points on the spectrum of political ideology alignment while also being available on the Nexis database. This framework of political ideology alignment of major news sources allowed for the investigation of political ideology in this study while avoiding relying on the subjectivity of the author in determining media outlet political alignment. A diverse set of outlets was thereby selected, capturing a range of perspectives on hydroxychloroquine in this period and ensuring the analysis considered ideological influences on how the topic was presented to the public in a way that is replicable and explicitly traceable. The news outlets chosen and their position on the Ideology of Online News Media scale are as follows:

- Far-right (2–3 on the Ideology of Online News Media scale)
 - ☐ Daily Caller
 - ☐ The Spectator
 - ☐ Newsbusters
- Right (1–2 on the Ideology of Online News Media scale)
 - ☐ Fox News
 - ☐ National review
- Centre-right (0.5–1 on the Ideology of Online News Media scale)
 - ☐ National interest
 - ☐ Realclearpolitics
 - ☐ Forbes
- Centre (−0.5–0.5 on the Ideology of Online News Media scale)
 - ☐ Associated Press
 - ☐ Politico
 - ☐ CNN
- Centre-left (−0.5 -> −1 on the Ideology of Online News Media scale)
 - ☐ New York Times
 - ☐ Newsweek
 - ☐ The Guardian
- Left (−1 -> −2 on the Ideology of Online News Media scale)
 - ☐ New Yorker
 - ☐ Salon
 - ☐ Quartz

- Far-left (−2 -> −3 on the Ideology of Online News Media scale)
 - The Nation
 - Vanity Fair
 - Prospect

2.4. Eligibility

Results from the search were eligible for inclusion in the study if they had relevance to the topic: they focused on hydroxychloroquine in the context of an alternative COVID-19 medicine and the public awareness and conspiracism around that topic. If the medium made only a passing reference to the topic or was fundamentally about something else, then it was excluded. News media were required to be in English and published within the temporal window relevant to this study (17–30 March 2020).

2.5. Data Extraction and Analyses

Following the search criteria above, 228 news articles and transcripts were downloaded from Nexis. Deselection followed as described below in the PRISMA 2020 flow diagram [28]. A total of 128 pieces of media content were included for data extraction and analysis.

Data analysis followed Graneheim et al.'s [29] approach involving keyword searching and manual coding performed with a priori codes using software Nvivo 14™. A priori codes were developed from a review of existing studies on hydroxychloroquine and similar misinformation and news media analyses, as well as from theory around misinformation such as fearmongering and ingroup/outgroup theory. Secondly, media content was indexed and read through for data exploration. Thirdly, content was manually coded qualitatively over three iterative rounds of coding. This manual coding allowed for the capture of the nuances of media framing. Fourthly, that content index informed data focusing using analytic tools within Nvivo 14™ and integrated learning from the wider academic literature. In the fifth stage, Nvivo 14™ data visualisation tools were utilised to test and refine themes found within the data to ensure these emergent themes were grounded in both the data and the broader literature. Sixthly, data codes were transferred to R version 4.4.1 to enable greater visual and statistical analyses. The transition of codes from Nvivo to R allowed for a deeper examination of the distribution of themes across media outlets and political ideologies, facilitating analysis of potential ideological asymmetry as outlined in research question 2. Finally, code distribution per media outlet and political alignment was analysed to assess potential asymmetry per RQ2. Codes are presented with representative verbatim quotes. All analysis was conducted by one author (RD).

The analytical tools used in this study allowed for intentional, replicable analysis while still providing the flexibility needed for identifying themes in the content and exploring manipulative mechanisms present in the data. Nvivo is an excellent tool for qualitative coding, enabling the identification of recurring themes within the media content that aligned with the a priori codes derived from the literature such as ingroup/outgroup dynamics and the framing of misinformation to general audiences. The iterative nature of this process ensured that emergent themes were refined across multiple rounds of coding. Expanding on this qualitative coding, R then allowed for the use of powerful statistical and coding tools to manipulate and visualise the data. This process provided insight into code distribution across media outlets, which in turn allowed for a clear, replicable quantitative comparison of ideological biases in media reporting on hydroxychloroquine in the relevant period.

3. Results

3.1. Articles Included

Eligible news articles and sources identified for this study were not evenly distributed among the media outlets chosen for inclusion (Figure 1). Several media outlets searched had no eligible content, including many of those on the left and far-left. This could reflect an asymmetry in coverage of hydroxychloroquine along the political spectrum, but may

simply mean that Nexis was missing content which could have met the eligibility criteria for inclusion (an unlikely but possible explanation—Nexis has relatively extensive broad coverage of these outlets). Alternatively, it could mean that some of the media outlets chosen for inclusion considered coverage of hydroxychloroquine out of their scope. CNN and Fox News are highly represented, which could reflect the higher levels of coverage possible through cable television media. Below please note the PRISMA flow diagram showing the search process by which these results were developed (Figure 2).

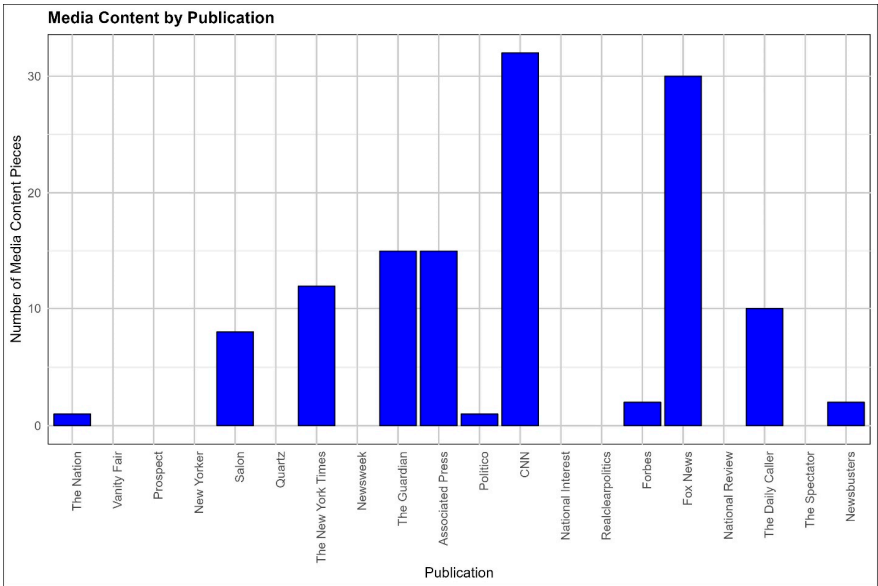


Figure 1. Bar chart showing the number of media pieces included per media outlet (x-axis organised by position on the political ideology alignment spectrum (far-left → far-right)).

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

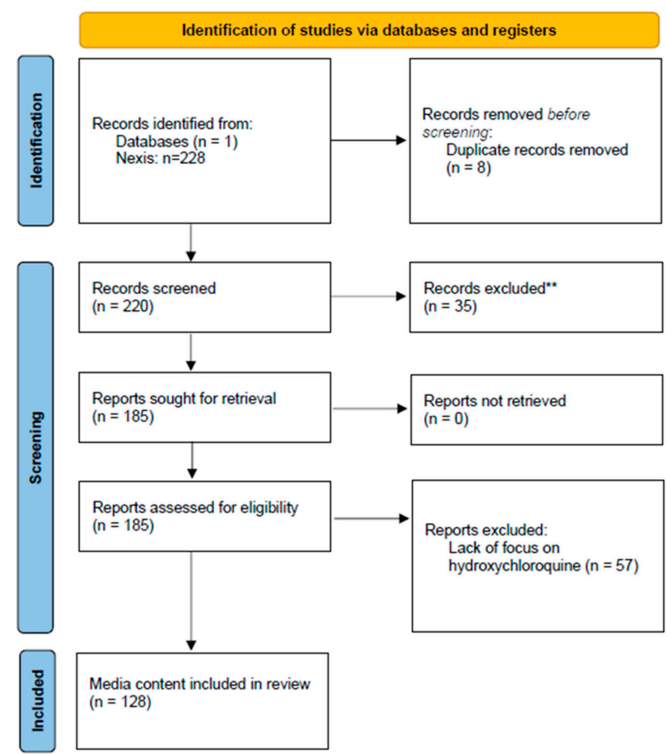


Figure 2. PRISMA 2020 flow diagram of media content identification.

3.2. Codes

Codes used in the analysis are shown in Table 1, which details explanations of the code, an example from the content, and the number of references coded throughout the content. For further information on the data utilised for this study, a supplementary file is available as noted on page 23.

Table 1. Analytical codes.

Code	Explanation	Example	Refs.
Realist framing	Framing events in a realistic light, highlighting both sides	“There is little evidence that the drugs work to treat coronavirus, although clinical trials are underway to find out” (Salon 2020)	107
Highlighting of elites	Amplifying elite speech	“thanks to global publicity—including from Donald Trump” (Guardian 2020)	98
Conspiracism	seeking/positing alternative explanations	“What if there’s already a cheap and widely available medication that’s on the market to treat the virus? . . . It’s called Chloroquine.” (Fox News 2020)	93
Politicisation	Inflaming tensions by blaming political opponents	“I mean the President shouldn’t mention any promising drug because, heaven knows, we don’t need any hope these days?” (Fox News 2020)	85
Anecdotal evidence	Presentation of anecdotal evidence or use of personal observation as evidence	“‘Bachelor’ star Colton Underwood . . . also revealed the medication he is taking to help recover from COVID-19. . . hydroxychloroquine” (Daily Caller 2020)	60
Optimistic framing	Framing events in a hopeful light, highlighting poor evidence as promising	“Clinical Trial Raises Hopes That Malaria Drug Could Be Coronavirus Cure” (Daily Caller 2020)	58
Public health messaging	Messaging from legitimate public health sources	“the FDA reiterated in a statement hours after Trump’s remarks that there are ‘no FDA-approved therapeutics or drugs to treat, cure or prevent COVID-19.’” (Associated Press 2020)	55
Pessimistic framing	Framing events in a dismissive light, highlighting the poor evidence around hydroxychloroquine	“Highlight: There is little evidence chloroquine and hydroxychloroquine are effective” (Forbes 2020)	42
Fearmongering	Inflaming anxiety and fear around hydroxychloroquine and the COVID-19 pandemic	“HANNITY: Dr. Oz, he warned this morning on ‘FOX & Friends’ that, quote, New York City is a war zone” (Fox News 2020)	26
Ingroup/outgroup	Praising those who tow the party line, shunning those who oppose party worldview	“Calling All Patriots: If You Had Coronavirus, Would You Take Hydroxychloroquine Or Other Experimental Drugs?” (Daily Caller 2020)	25

3.3. Realist Framing

Realist framing—demarcated by media attempting to take an objective, realistic, multi-faceted perspective towards hydroxychloroquine—was the most common code in this content. Much of the content herein is realist in nature and attempts to highlight both the positive and negative aspects of hydroxychloroquine as a potential treatment

for COVID-19 undergoing active experimentation. As realist framings are prevalent and consistent in this content, this could imply that the less prevalent framings (for example, conspiracist) are more persuasive. It also highlights the insufficiency of this framing in responding to misinformation, as the prevalence of realist framing in this period failed to hinder conspiracism. It is also significant to note that although the most common code in this content, realist framing is nearly absent entirely from the right-wing and far-right content—almost exclusively present in left-wing and centrist outlets (Figure 3).

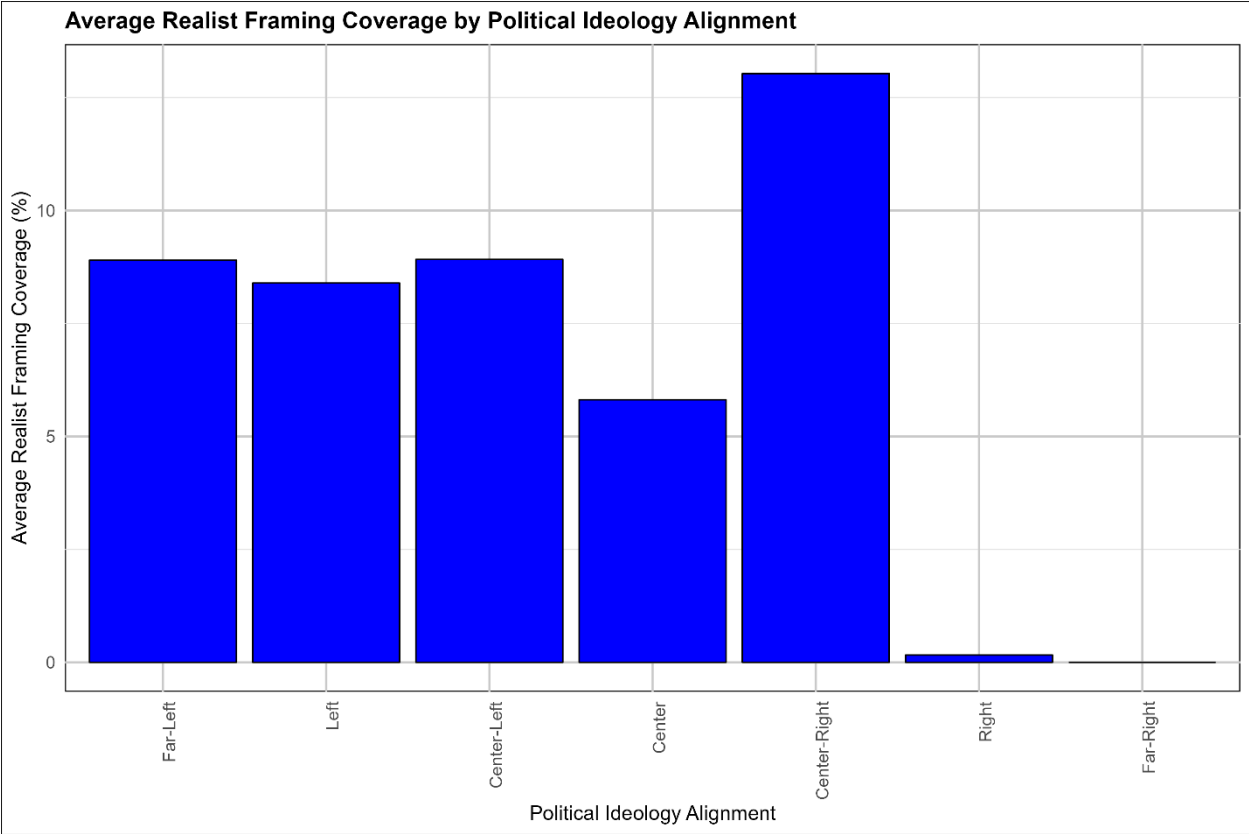


Figure 3. Bar chart showing average realist framing coverage (measured as % of wordcount coded thereto) by political ideological alignment (The percentages in this figure and the following similar figures refer to an estimate of how much of the wordcount in a given article is taken up by the relevant framing. These percentages will not add up to 100, as the media content herein contain a variety of background content alongside relevant discussions of hydroxychloroquine).

3.4. *Highlighting of Elites*

A large portion of all included studies named a political elite (most often Trump) in the headline. This headlining frames the entire article as an extension of the elite’s speech and serves to amplify and legitimise their speech. Beyond the headlines, the repetition of elite speech and portrayal of their perspectives served to draw public attention to the speech and extend their audience considerably. This extended across the political spectrum in this content (Figure 4).

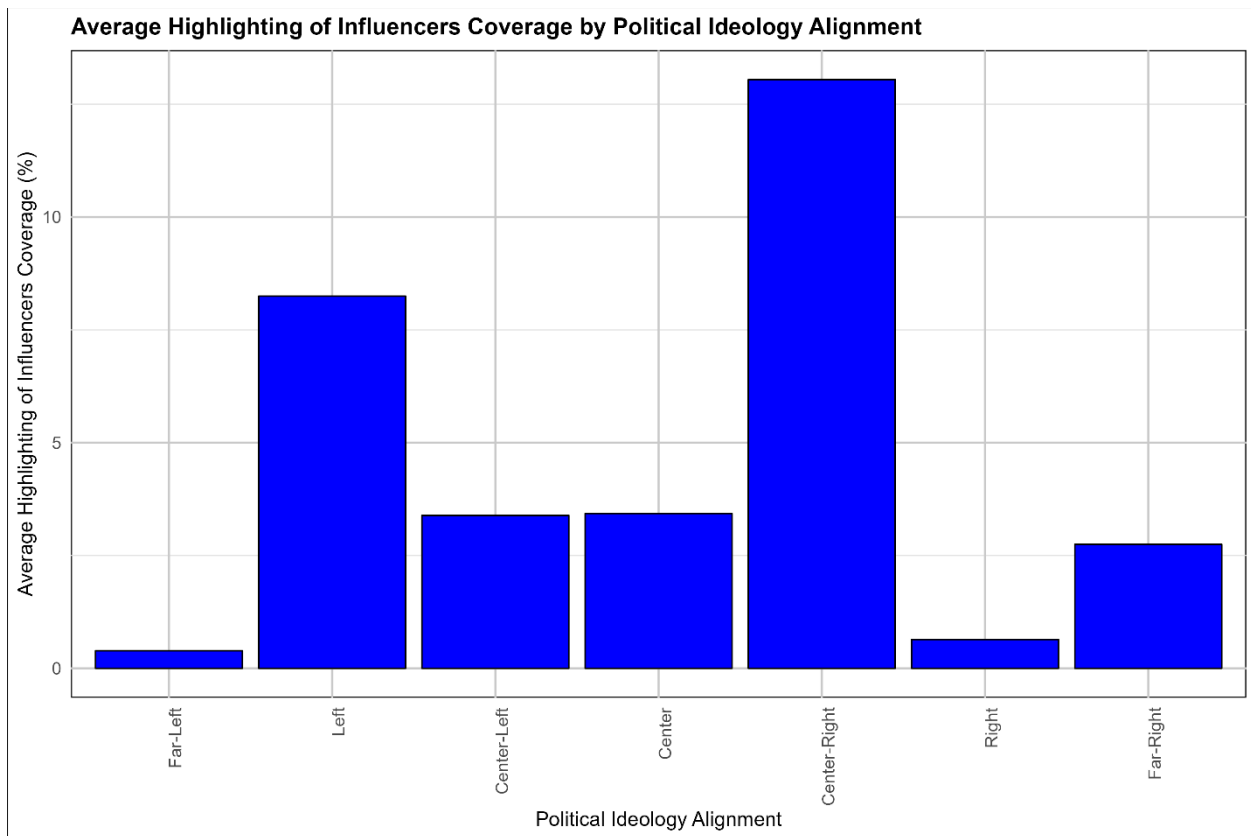


Figure 4. Bar chart showing average highlighting of elites coverage (measured as % of wordcount coded thereto) by political ideological alignment.

3.5. Conspiracism

Conspiracism was widely coded in this content, skewing to the right politically (Figure 5). It is distinguished by the use of larger narratives to explain and mislead audiences, giving viewers the sense that they are ‘in on it’ and special for knowing the ‘hard truth’. The emotional appeal therefrom can be very powerful. This framing is among the most complex forms of manipulation present in this content. Perhaps the most noteworthy example of manipulation in this study came from Tucker Carlson Tonight [30]:

CARLSON: Now, keep in mind, this isn’t a tax bill they’re lying about. It’s a potentially life-saving medicine that we are desperately trying to evaluate in the middle of a global pandemic. And it’s not just CNN that’s doing it. Consider this Bloomberg news story from last night, “Chloroquine no better than regular coronavirus care, study finds.” That was the headline and it sounds definitive.

Why are we wasting our time with this crap? It doesn’t work.

But take a look at the details inside the piece. The story turns out to be based on an article in the journal of Jiaxing University in China. The Chinese study in the article considered a total of just 15 patients—15—who have been given chloroquine. It seems like an awfully small sample for a legitimate study. And in fact, the story does concede that was.

“The results of the study weren’t statistically significant. Researchers concluded that additional studies using larger numbers of patients are needed.” In other words, the story doesn’t really tell us anything.

So why is Bloomberg News writing about it? Later in the story, we learned that in contrast to the protocol in that tiny Chinese study, doctors here in the West are

administering hydrochloroquine with antibiotics, and that combination, the two of them hydroxychloroquine and antibiotics appears to be effective.

And that raises the same question once again. What exactly is the point of Bloomberg's so-called news story? The way the Chinese used the drug had nothing to do with the way we are using it here. The study is irrelevant. Writing about it doesn't illuminate anything. It misleads.

That's the point. They're manipulating you. This is what happens when science becomes political. Suddenly, people start lying, and you can't trust anything you hear. That's a very dangerous thing to do at a moment like this. The truth is essential.

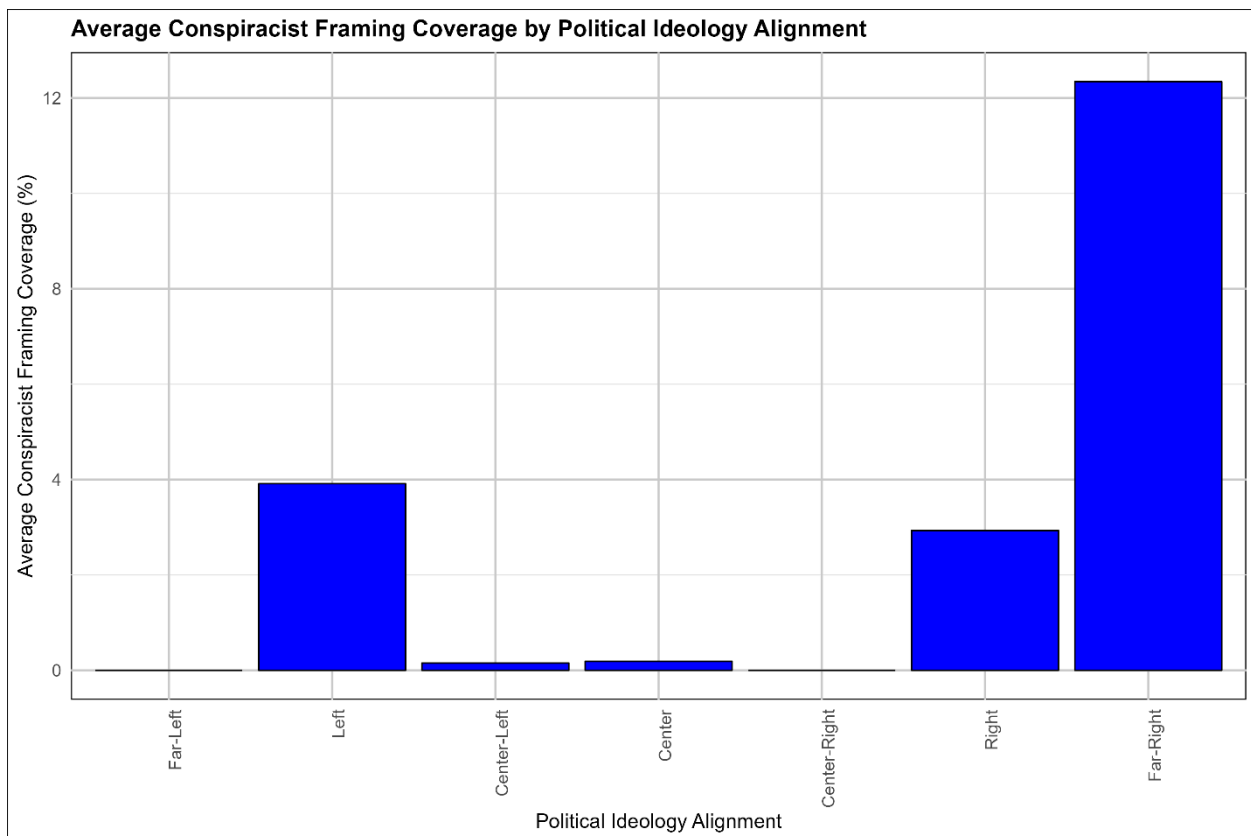


Figure 5. Bar chart showing average conspiracism coverage (measured as % of wordcount coded thereto) by political ideological alignment.

This example stands out for the veneer of scientific language and analysis presented to the audience. Here, Carlson suggested that the mainstream media are lying to viewers, attempting to manipulate them with politicised improper science. Carlson even highlighted an example of poor scientific media portrayal from CNN and Bloomberg News, two centrist-aligned news sources, in which a study with insufficient participant numbers and lack of statistical significance is reported on as if it is confirming. Although his rejection of that study as not 'really tell[ing] us anything' neglects to discuss the nuance of null hypothesis testing and interpretation thereof, he provides a convincing argument that the study in question is insufficient for the interpretation drawn by the news media. Yet underneath this veneer, manipulation can be seen. In this example, Carlson appeared to be warning viewers about mechanisms of misinformation. This is an example of Accusation in a Mirror, a propagandist tactic well established in the literature on propaganda and misinformation [31], in which a speaker accuses their opponents of doing what they and their own party are planning to do. In this example, Carlson accused CNN and Bloomberg News of doing what he and Fox News are doing: using poor scientific evidence

to support politicised claims. This simultaneously delegitimises other opposing media sources, legitimises Carlson and Fox News (because they are warning viewers of these manipulative tactics), and teaches the viewer how to reject information incoming from other news sources that may contradict the worldview promoted by Fox News.

3.6. Politicisation

Politicisation occurred widely in the content, in a horseshoe-shaped distribution with greater prevalence on the political extremes at both ends of the spectrum (Figure 6). This politicisation typically looked like each side pointing fingers at the other side for the pandemic, and the use of hydroxychloroquine rhetoric to support existing political disagreements. In this context, politicisation refers to a particular manipulative tactic present in the content wherein the media outlet comments on a piece of news as *too political* or otherwise politically motivated or portraying one political side unfairly. These accusations distract from and conflate news stories, serving to appeal emotionally to audiences and encourage ingroup mechanisms.

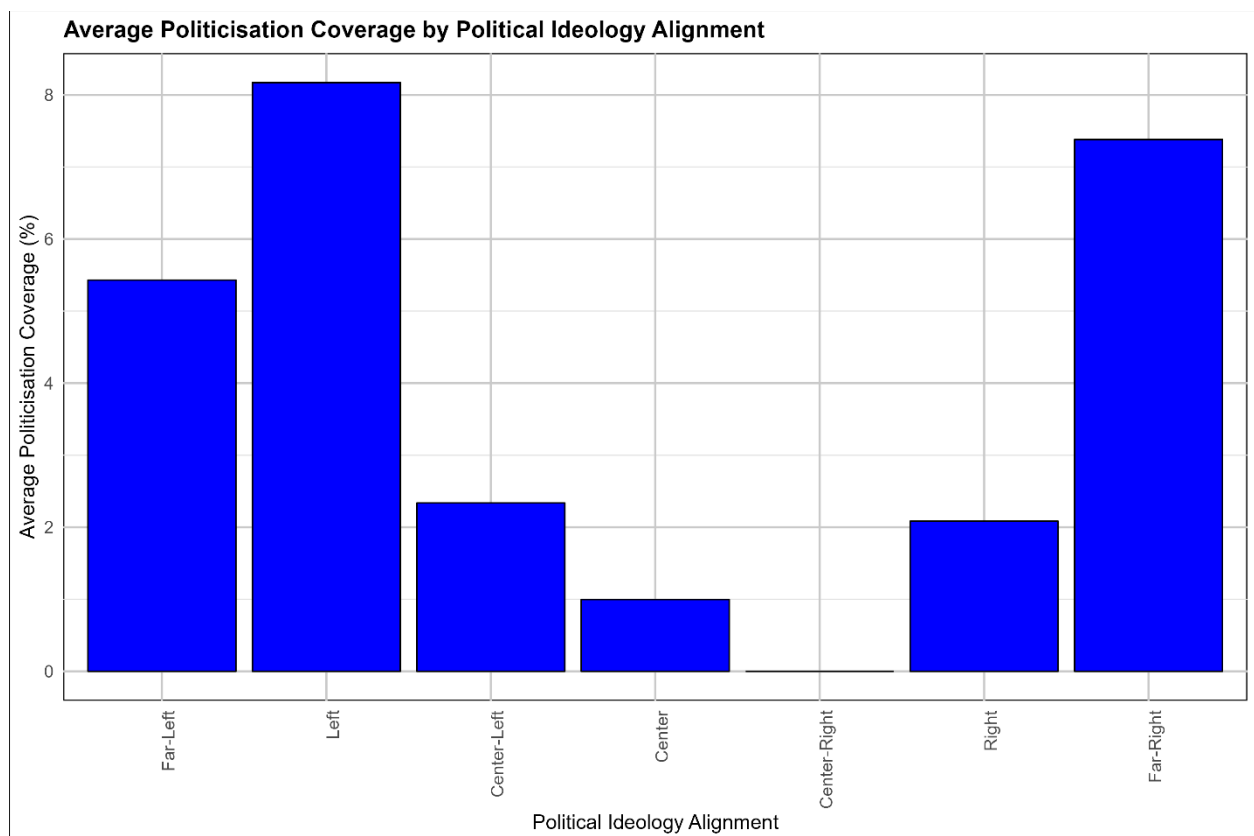


Figure 6. Bar chart showing average politicisation coverage (measured as % of wordcount coded thereto) by political ideological alignment.

Within some of the Fox News content in this study are examples of television hosts exemplifying and thereby teaching intransigence to the viewer—an extreme and noteworthy form of politicisation. This looked like the host walking the viewing audience through how to reject evidence that goes against the conservative ingroup worldview. A key example came from the Ingraham Angle [32]:

INGRAHAM: Admiral, I need to ask you about this brand new study... it is clear from this... that this hydroxy-azithro combo seems to be working. So why are doctors and nurses and pharmacists today getting mixed messages... on the need for a controlled study?

ADMIRAL BRETT GIROIR, WHITE HOUSE CORONAVIRUS TASK FORCE: In order to be absolutely certain that the drug combination works or one of the other drugs work, sure, you need a controlled study. . .

And as you described, the study that was published tonight was on 80 patients. It was not controlled. But it’s again another piece of suggestive evidence. . .

INGRAHAM: But this is where you see this—the tension, Admiral, in the medical community. People are not sure what to do. . . . And I’m talking to infectious disease doctors every day. . . This is their go-to combo in most patients. Period.

In this example, Laura Ingraham began by framing the question as if the evidence is clear that hydroxychloroquine works as a treatment for COVID-19, then asked why the administration is insisting on further testing. The admiral then corrected her, that controlled studies are required with larger participant pools to be certain of efficacy. Ingraham then framed this response around confusion in the medical community, suggesting to viewers illegitimacy and mixed messaging on the part of the medical establishment, which could prime them to disregard legitimate public health messaging. She ended by mentioning her anecdotal evidence talking with doctors who use hydroxychloroquine to treat COVID-19, even ending her comments with the word ‘period’. This together suggests that anecdotal evidence is significant and confirming and needs no further controlled testing. The viewer is not only told that Laura Ingraham’s anecdotal evidence supersedes the controlled testing called for by the medical establishment, but shown precisely how to counter and reject arguments from the medical establishment calling for further testing.

3.7. Anecdotal Evidence

High levels of anecdotal evidence were present in the content across the political spectrum (Figure 7). This presentation of evidence is particularly fitting for news media, as it allows for human interest storytelling methods and greater emotional appeal to audiences. This coding highlights how poor evidence is used in ways that insufficiently inform the audience of how poor the evidence being presented actually is.

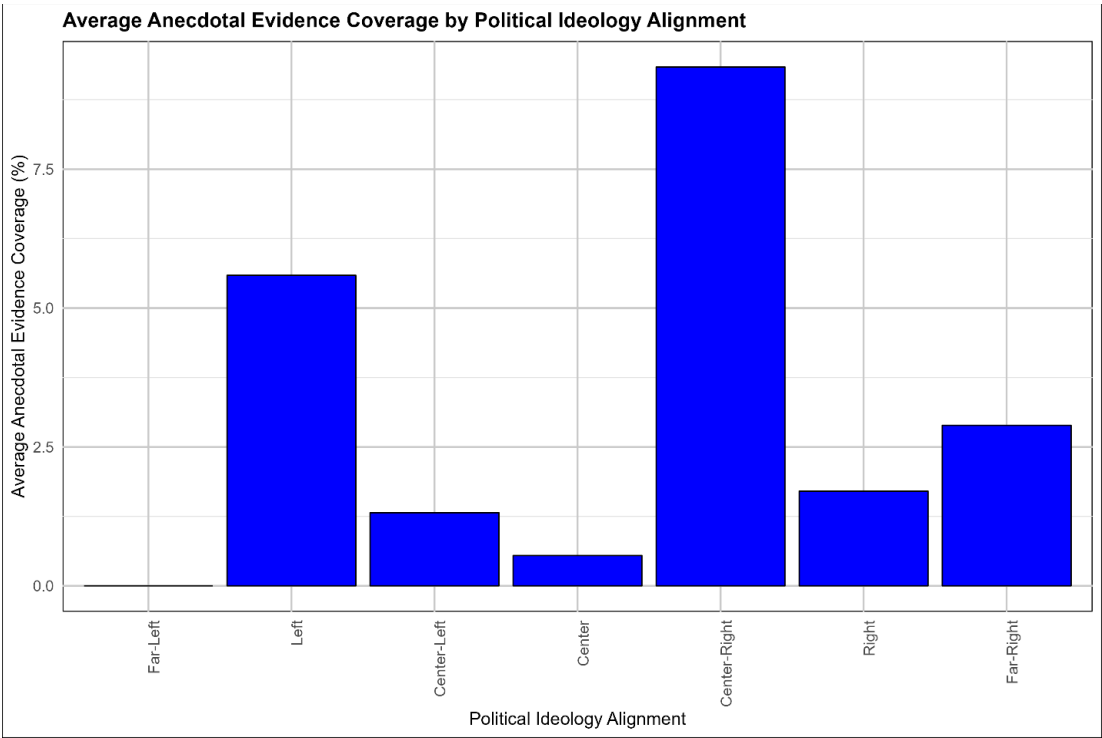


Figure 7. Bar chart showing average anecdotal evidence coverage (measured as % of wordcount coded thereto) by political ideological alignment.

3.8. Optimistic Framing

Optimistic framing was disproportionately found in right-wing and far-right content, absent from left and far-left content (Figure 8). The optimism here is both misleading based on the evidence available for hydroxychloroquine as a treatment for COVID-19 at the time and allows for greater emotional appeal to audiences. Fox News hosts often framed discussions around the use of hydroxychloroquine as a question of ‘how far are you willing to go to protect your family?’ or ‘are you going to let the government regulators stop you from saving your child’s life?’. This emotional appeal and connection to motherly nurturance of sick children or patriarchal protection of the family served to numb audiences to the poor evidence base around hydroxychloroquine and instead portray the issue as one about freedom, independence, and doing whatever it takes to protect your family from imminent danger. This scenario is exemplified in the following quote from an interview [33] between Sean Hannity of Fox News and Dr. Jerome Adams, the U.S. Surgeon General at the time:

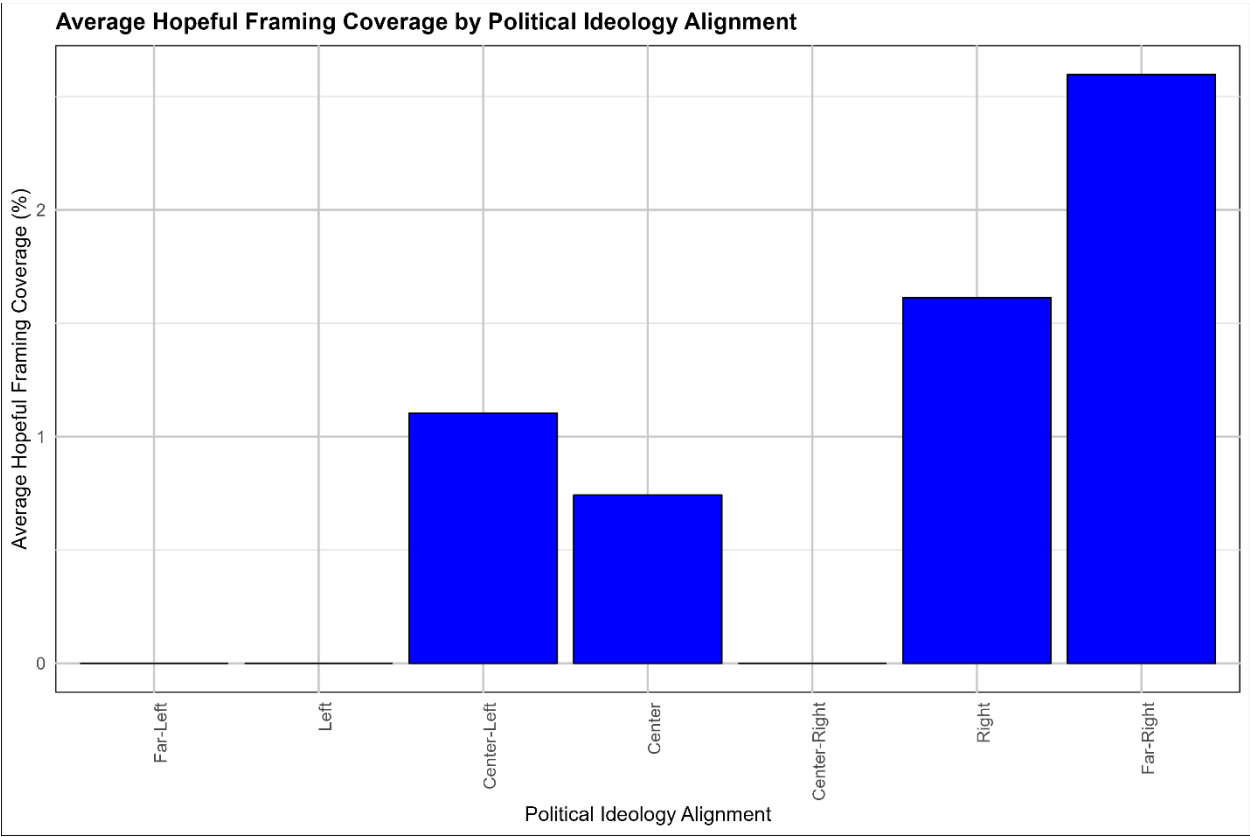


Figure 8. Bar chart showing average optimistic framing coverage (measured as % of wordcount coded thereto) by political ideological alignment.

HANNITY: I’m almost out of time. Would you take hydroxychloroquine and azithromycin if it was you or a family member? Real quick, I only have about 15 s.

ADAMS: If it was me or—if it was me or a family member and they were in the hospital in the ICU, I would try anything that I could to help keep them alive, and that’s what we want to get people—

HANNITY: So would I. ADAMS:—that opportunity to try to keep their loved ones alive.

HANNITY: Sir, thank you for all your doing and your entire task force.

In this quote, a variety of manipulative mechanisms are at play. Hannity framed the question as rushed and running out of time so as to demand a brief answer, and even

interrupted Adams’ answer to best shape the answer into the framing desired. This gave the guest no opportunity to respond, and nullified any possible sentiment previously mentioned in the guest’s appearance that could be seen as anti-hydroxychloroquine or otherwise pessimistic or calling for additional evidence, such as this line from Adams immediately before the above quote: “We—we are hopeful that these medications will work, we’ve heard some powerful anecdotal stories, but we want to make sure we’re tracking.” This treatment of guests—ending a guest appearance with a rushed question about whether the guest would take it or use it on their sick family members, with a cut-off response, was repeated in the Fox News content, and serves as a powerful example of the manipulation present in their content.

3.9. Public Health Messaging

Public health messaging (Figure 9) during this time period largely came from the mouth of Dr. Anthony Fauci, a lead member of the White House Coronavirus Task Force under Donald Trump and the Chief Medical Advisor to the President of the United States under Joe Biden. Therefore, Dr. Fauci came to represent the public health apparatus in the Trump White House and became widely perceived as the chief medical expert on COVID-19 for the United States with high levels of public influence and legitimacy [34]. When Fauci came to public disagreement with President Trump on the issue of hydroxychloroquine, this represented a far greater schism than simply between the two individuals. Trump was the sitting President, the leader of the Republican Party, and a charismatic, widely loved political elite figure among conservatives [35,36]. Due to the complexities involved in Dr. Fauci’s symbolic position as representing the public health apparatus in the Trump White House, the split between the two men catalysed and deepened an existing rift between public health and conservatives whose impact is ongoing.

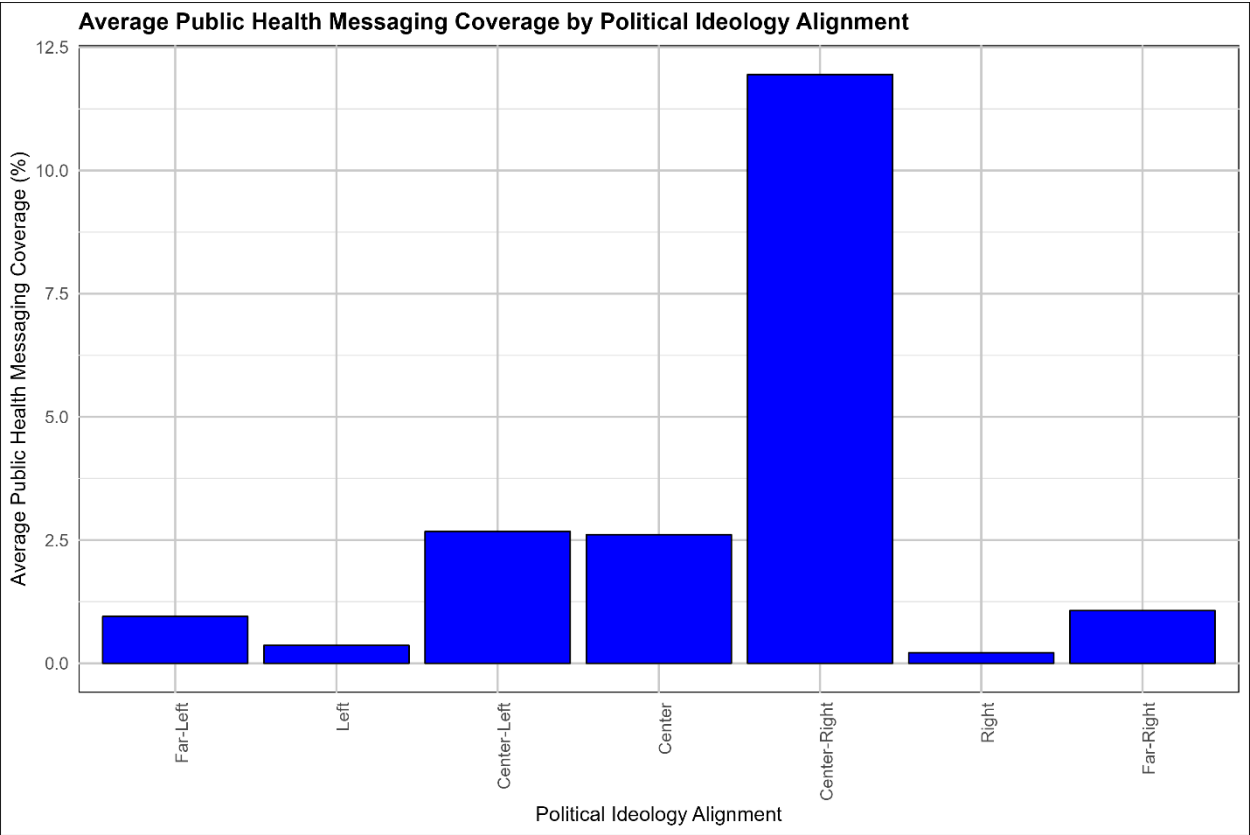


Figure 9. Bar chart showing average public health messaging coverage (measured as % of wordcount coded thereto) by political ideological alignment.

Overall, the public health messaging present in the content of this study was overly focused on the promise of hydroxychloroquine while emphasising the need for further study. One example from a centrist-aligned outlet, CNN, read “Dr. Anthony Fauci, the nation’s top infectious disease expert, has said the drugs may be effective against the novel coronavirus. . . ‘I’m not disagreeing with the fact anecdotally they might work, but my job is to prove definitively from a scientific standpoint that they do work,’ said Fauci” [37]. The reporting here began by highlighting that the drugs may be effective, and that it works anecdotally, but did not move on to describe how it could be proven from a scientific standpoint. It read as hopeful and positive about hydroxychloroquine, even while explicitly written as a warning against taking hydroxychloroquine. This ‘halfway’ approach that described hydroxychloroquine as ‘promising’ and acknowledging the extant promising (although insufficient) evidence, served to functionally promote hydroxychloroquine to public audiences. Public health scientists can easily distinguish between anecdotal evidence and more substantial evidence, and can acknowledge the nuances of scientific evidence—but the public are not trained scientists.

3.10. Pessimistic Framing

Pessimistic framing was spread across the centre and left of the spectrum and entirely absent in the right wing (Figure 10). It was also less prevalent overall in the content than optimistic or realist framing, highlighting the lack of pessimism in this period related to hydroxychloroquine as a treatment for COVID-19. This framing primarily showed in the content as politicised attacks on President Trump from the far-left, with much of the other codes mixed with realist coding in the political centre.

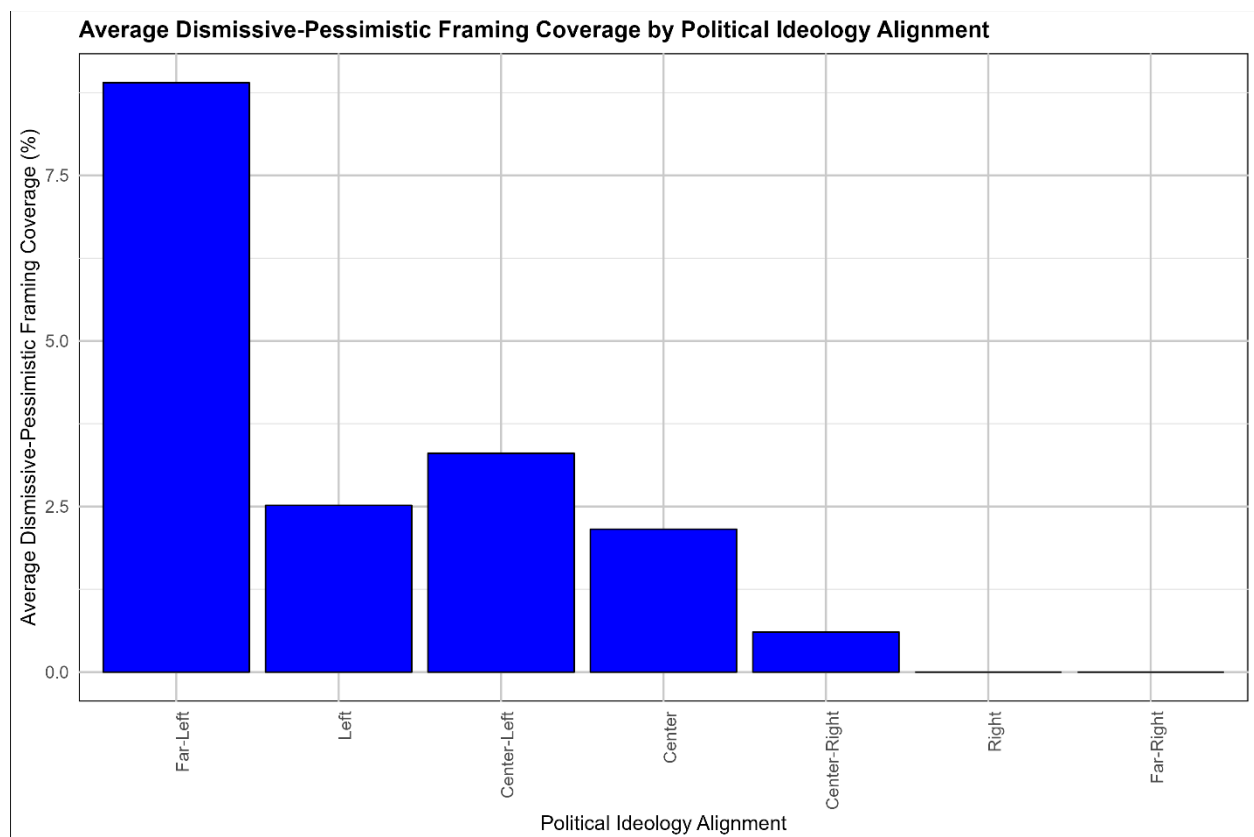


Figure 10. Bar chart showing average pessimistic framing coverage (measured as % of wordcount coded thereto) by political ideological alignment.

Several of the media pieces included for analysis are direct fact-checks from different outlets addressing Trump’s false claim that the FDA approved hydroxychloroquine for

treating COVID-19 [38,39]. This false claim was addressed and attempted to be countered by a wide variety of politically centre/left media outlets in the content in this study. Alternatively, Fox News reported that the FDA did approve hydroxychloroquine, and that other outlets were being pessimistic in their coverage of it, saying of the Washington Post: “You can tell they will be heartbroken if the drug actually works.” [30]. This phenomenon highlights the right-wing media as aware of the rest of the media ecosystem as more pessimistic regarding hydroxychloroquine as a treatment for COVID-19 and actively mocking them for their pessimism.

3.11. Fearmongering

Throughout much of the content but particularly prevalent on the more extreme left and right politically (and skewing right), fearmongering framing could be seen (Figure 11). This framing allowed for emotional appeal and capturing the attention of viewers, stoking the flames of fear and polarisation in this time. An example of this framing came from The Nation, a far-left-aligned outlet: “The Trump administration’s response to this crisis continues to be a combination of ineptitude and criminal lack of concern for the most vulnerable. Some particularly awful highlights, on the malign-ineptitude front: The president continues to peddle snake-oil cures that his own infectious diseases specialists then have to pour cold water on.” [40]. This framing allowed for more biased language, greater hyperbole and exaggeration. Another example comes from Fox’s Ingraham Angle [41]:

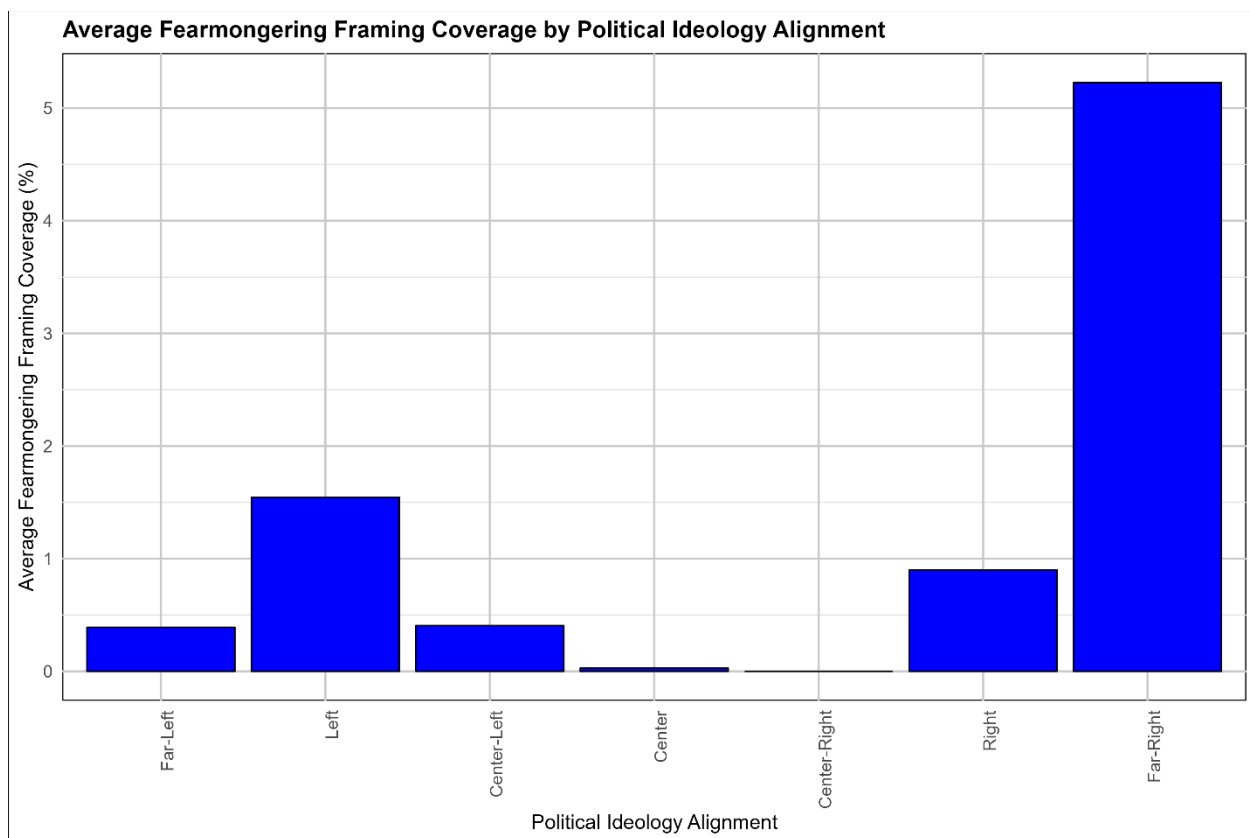


Figure 11. Bar chart showing average fearmongering coverage (measured as % of wordcount coded thereto) by political ideological alignment.

OZ: And so we’re going to want a sophisticated program like they had in South Korea, we can do it.

INGRAHAM: And Dr. Oz, so you’re saying that... the federal government should work with big tech to track Americans’ movements and then punish them if they are not obeying what the government is saying.

OZ: No, no, we can't do what they did in South Korea obviously. That's not who we are as a people. . . most Americans wouldn't want that.

In this example, the guest (Dr. Oz) discussed the uses of a sophisticated, private-public partnership to trace the spread of the virus. In response, Laura Ingraham framed the issue as a dystopian system of tracking citizens' movements and punishing people for 'not obeying'. She twisted the idea to frame it as scarier and more authoritarian.

3.12. Ingroup/Outgroup Mechanisms

Ingroup/outgroup mechanism codes skewed to the far-right in this content (Figure 12). One powerful example of the usage of ingroup/outgroup mechanisms came from an Ingraham Angle [42] clip from Fox News:

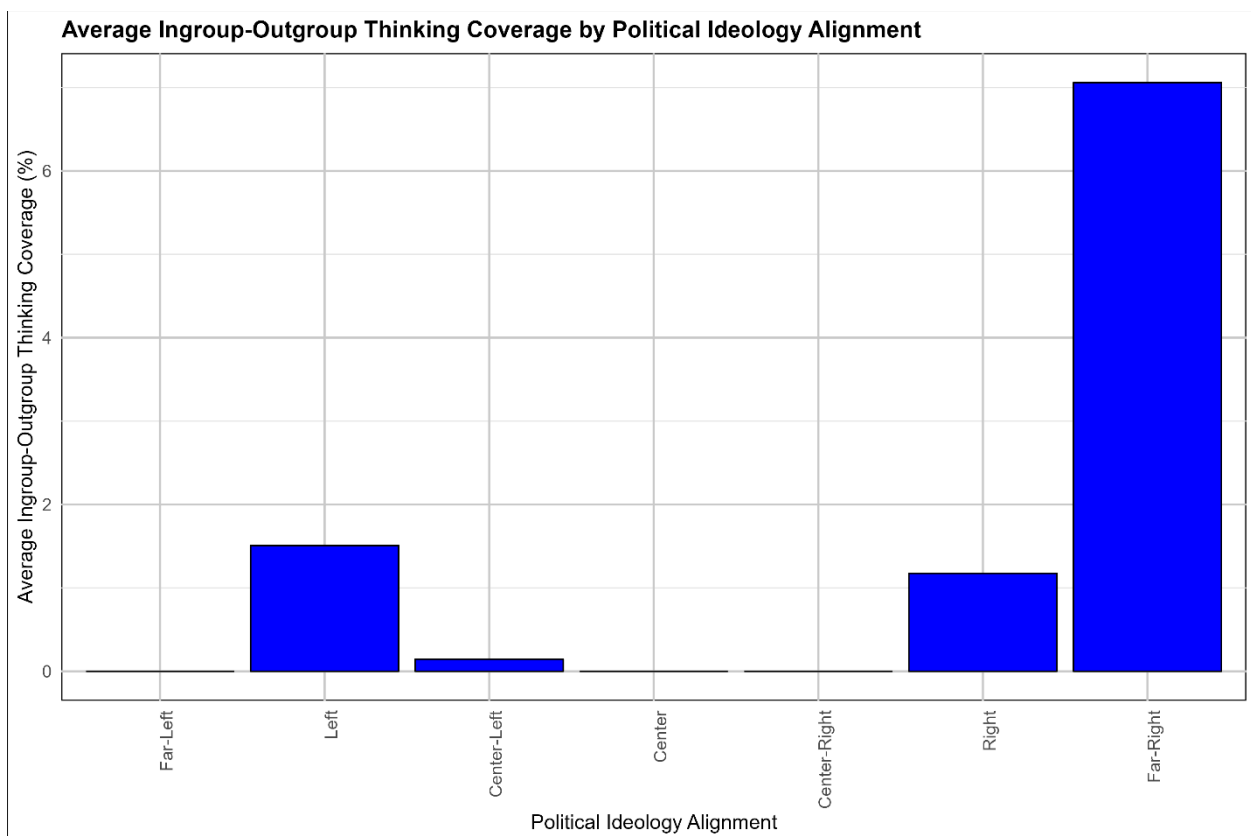


Figure 12. Bar chart showing average ingroup/outgroup thinking coverage (measured as % of wordcount coded thereto) by political ideological alignment.

SANTILLI: It obviously worked on me. It's worked on other people as we've seen around the country and quite frankly, around the world. . . for any person to not want this to work simply because President Trump mentioned it, it's truly unamerican and they should be ashamed of themselves. This is about saving lives, stopping the suffering and getting through this together as a nation. COVID-19, it's not a partisan issue.

INGRAHAM: Jim, I could not have said it better myself.

Santilli claimed to have taken hydroxychloroquine to treat his own COVID-19 and dramatically recovered thereafter. The host (Ingraham) and guest (Santilli) reinforce each other's statements that those against the use of hydroxychloroquine are unamerican, should be ashamed, and are working against the interests of the nation. This informs viewers that those who do use hydroxychloroquine are to be praised, and those who are against its usage must be cast out from the ingroup as disloyal.

4. Discussion

Coverage differed across the political spectrum. As outlets became more radical, politicisation increased notably. The further-left outlets were dismissive and negative towards hydroxychloroquine whereas the further-right outlets were hopeful and positive, even to the extent of manipulating and misinforming viewers. Fox News, the Daily Caller, and Newsbusters were all crucial to the propagation and dissemination of the relevant conspiracism. The alternative perspectives on the left were less significant, had fewer examples of manipulative tactics in the content, included fewer conspiracist framings, and took a more evidence-based view of hydroxychloroquine.

4.1. Contribution to the Wider Literature

The results of this study help to explain the pendulum effect noted by Sattui et al. [9] by highlighting the role of the media in propagating and disseminating misinforming and inflammatory speech by conservative elites, as well as promoting anecdotal evidence and (in the case of Fox News and the Daily Mail) manipulating their audiences. Further, the results reinforce existing findings in the wider literature that identify the American far-right and Fox News as manipulative and promoting of conspiracism. Within the wider literature on COVID-19 conspiracism and hydroxychloroquine, these results provide novel insight into the politics involved. COVID-19 conspiracism is intensely political, benefiting the American far-right and with media outlets using manipulative tactics in their presentation of the material to public audiences.

This study contributes to wider discussions in the scientific literature on the connections between political ideology and misinformation, media and the spread of misinformation, and misinformation as a coping mechanism in uncertain times. It is well established in the literature that political ideology plays a significant role in the acceptance and internalising of misinformation [43], particularly during crises like the COVID-19 pandemic [44]. Studies have found individuals with conservative ideologies to be particularly susceptible to vaccine misinformation and conspiracism related to the pandemic [45,46]. This study builds on the existing literature identifying political ideology as significant regarding conspiracy theories around the origins of COVID-19 to suggest that it might also play a role in conspiracy theories around the treatment of COVID-19 [47]. The patterns identified here connect to the theory of motivated reasoning, wherein individuals engage with information that reinforces their pre-existing beliefs and identity—rejecting contradictory evidence [48].

This study participates in discussions around the role of different media forms in the contemporary information age and how alternative media sources play into the dissemination of misinformation. While it is increasingly considered common knowledge that misinformation and conspiracism have grown since the emergence of social media [49], this study shines a light on the continued impact of traditional media forms in promoting and perpetuating misinformation arising initially from social media space. Furthermore, this study pushes back against the idea that traditional media are not contributors to misinformation, with some studies suggesting that consumers of traditional media are less likely to be misinformed [50,51]. The results of this study indicate that traditional media continue to play an important role in shaping public perception of major events and providing opportunities for partisan elites to influence viewers. Therefore, traditional media should be seen as symbiotic with social media, adopting viral strategies and amplifying online misinformation [52].

4.2. The Purpose of Public Health Messaging

A core learning of this content analysis is the insufficient persuasiveness of traditional public health messaging relative to the manipulative tactics utilised in both news media and misinformation. If the public health apparatus is constrained by ethical/moral desires for ensuring autonomy in individual decision making and desires to provide only the most unbiased, objective possible messaging, such tactics will therefore struggle when brought to bear against those who have no such qualms and utilise far more powerful tactics like

manipulation, emotional appeal, lying, etc. This creates an inherent inequality in the tools available to legitimate attempts to inform the public (i.e., public health messaging) relative to those seeking to misinform or simply seeking to provide eye-catching, emotionally appealing content for viewers (i.e., news media). Although this study does not directly analyse public opinion, the ramifications of highly persuasive news media content when reporting on scientific studies and public health can be devastating, as is the case when discussing the 17,000 unnecessary deaths related to improper usage of hydroxychloroquine to treat COVID-19 worldwide.

4.3. Political Asymmetry

A secondary research question for this study is “Did coverage of hydroxychloroquine in this time period differ substantially by political ideology?”. This research question has a clear answer arising from the data within this study. Yes, coverage of hydroxychloroquine differed substantially by political ideology. Ingroup/outgroup dynamics alongside the language, rhetoric, and tactics differing extensively between sides of the political spectrum points directly at the political right wing and particularly the far-right being integral to the creation and dissemination of the conspiracism and misinformation at the core of this study. This asymmetry and usage of manipulation stands to benefit a variety of groups. The media outlets benefit from reporting on inflammatory stories, fear, drama, and highlighting popular elites for higher engagement. Furthermore, outlets aligned with political extremes benefit from producing mistrust in other media outlets as it legitimises their content and reinforces audience loyalty. The elites benefit from reinforcing their support and loyalty via manipulation. The right wing and far-right benefit explicitly from accentuating division among the American public by reinforcing ingroup/outgroup mechanisms among their followers.

4.4. Limitations

This study has several important limitations. The narrow temporal window of the content analysed means there could be important trends and patterns missed, and that the themes derived from this content may appear different when seen in a wider temporal lens. This study focuses intently on a brief, highly relevant period, taking a purposefully limited, micro-perspective to best trace the early growth of the hydroxychloroquine conspiracy theory in real-time. Although that is the explicit purpose of the study, it is also inherently limiting. The qualitative nature of the data within this study, alongside the previously mentioned temporal limitations, serves to limit the validity and generalisability of the results. This study seeks to offer insight into the complex processes by which conspiracism and misinformation are generated and perpetuated by media ecosystems and the general public. While valuable for theoretical analysis and new insights, this study does not aim to provide a fully exhaustive or definitive answer. Furthermore, the limited media coverage of the content in this study limits the analysis. The media outlets were chosen for their fit in a framework on political ideological alignment and placement in the Nexis database, with other factors like their audience, impact, and legitimacy as tertiary factors. It is possible that in a wider context of media content, the patterns recognisable in this data may change or disappear. Additionally, the geographical restriction to the USA limits generalisability to other areas, although it appears similar phenomena occurred in Australia around Clive Palmer as a conservative elite spurring conspiracism around hydroxychloroquine. All research designs like this one, wherein search strategies are narrow and deselection is conservative risk missing important information within the data. Lastly, the content included in this study is not evenly distributed among the media outlets chosen, with several outlets having no representation. This could be due to missing content on the Nexis database (unlikely), or reflective of a difference in levels of coverage of hydroxychloroquine in the relevant period by news outlets with differing priorities and audiences.

4.5. Directions for Future Research

This study reveals the need for future research in several areas. First, more work is needed testing interventions against misinformation, particularly preventative interventions, as this study evidences both the slow reaction time of the public health apparatus and the contribution of the news media to quickly developing conspiracism and misinformation. Secondly, there is a need for better understandings of political asymmetry in misinformation, especially the role of the contemporary far-right. Third, research is needed investigating differences between how the general populace and those most vulnerable to misinformation respond to interventions, to gain better insight into the potential for backfire effects and specialist interventions for subgroups. Fourthly, this study highlights the need for greater understandings of the role of the news media in perpetuating and extending the reach of misinformation—as they benefit from methods of audience engagement that directly oppose public health efforts and explicitly benefit from promoting anecdotal evidence and inflammatory speech. Fifthly, further insight is needed into public opinion relating to hydroxychloroquine in this period and the surrounding timeframe to gain more concrete insight into the effectiveness of public health messaging in countering the misinformation and conspiracism studied herein. Sixthly, this study serves to exemplify the need for expanded insight into the manipulative and misinformative tactics and techniques present in the media. Such tactics are themselves worthy of considerable study as to best understand, learn from, and respond to them.

While this study primarily focuses on the role of political ideology and media in the spread of misinformation, further research could benefit from a semiotic analysis of the language used in media portrayals of COVID-19 conspiracies, particularly in conservative outlets like Fox News and the Daily Mail. Investigating the semiotic strategies employed—such as framing, metaphor, and symbolic representation—could offer deeper insights into how media content is constructed to resonate with specific ideological groups. By focusing on how language is used to shape and reinforce beliefs, future studies could help us better understand the persuasive power of media and their role in perpetuating misinformation. Understanding the semiotic mechanisms behind media narratives would complement the findings of this study and contribute to the growing body of literature on media manipulation in political contexts.

4.6. Recommendations

Following examination of the public health response to the rise of public awareness and subsequent conspiracism around hydroxychloroquine, it is clear the public health response in this case was insufficient, came too late and reactively, and engendered significant backfire effects. Therefore, the following recommendations relate to what could have been improved or done better in the public health response to hydroxychloroquine conspiracism:

1. Avoiding the pendulum effect: initial positive/hopeful framings of hydroxychloroquine followed by harsh/critical rejection of hydroxychloroquine served to drive conspiracists further into their conspiracism
2. Different messaging
 - a. Messaging that incorporates public understanding of evidence, not scientific understandings. It is clear throughout the content in this study that public health experts were too lenient on public understandings of anecdotal and insufficient evidence, and that the public were very fast to latch onto insufficient evidence and hold onto the hope that initial studies showed on hydroxychloroquine. This messaging could include elements intended to engender emotional responses or otherwise intentionally persuade public audiences.
 - b. Message framing that aligns with the worldview of the subgroup opposed to the traditional public health messaging. This will best allow for penetrating the powerful psychological mechanisms like ingroup/outgroup identities and give better potential for behavioural change outcomes.

3. Expectation of and preventative measures taken against conspiracism and desire for alternative medicine/treatments. These phenomena are normal and should be expected by authorities attempting to direct public behaviour in crises like the COVID-19 pandemic. Public information campaigns preventatively before conspiracism took hold around hydroxychloroquine could have substantially addressed that conspiracism better than reactively, as it happened.

5. Conclusions

News media portrayals of hydroxychloroquine contributed to the rise of conspiracism through a variety of mechanisms. First, media content amplified the voices of elites doing the misinforming, often highlighting elites even in the headlines of articles. These actors already had significant and committed followings, amplifying the power of insufficient evidence because their audiences are primed to believe them. This then gave those elites great influence over the discourse in this period. Next, the widespread use of hopeful/optimistic framings around hydroxychloroquine created a sense of optimism towards its use for treating COVID-19 across the political spectrum. The use of human-interest framings and widespread use of anecdotal evidence contributed to public misunderstandings of the scientific evidence available. Lastly, the high levels of politicisation—particularly from Fox News going so far as to teach intransigence to viewers—allowed for support of hydroxychloroquine to function as a powerful mechanism around which to gauge loyalty of conservatives to the ingroup, a key mechanism of radicalisation, polarisation, and conspiracism. Coverage of hydroxychloroquine in this period did differ substantially by political ideology, with right-wing and far-right actors being significant in the perpetuation and dissemination of misinformation.

Therefore, this study informs discussions of the birthplaces of misinformation and conspiracism, and how media coverage of this phenomena may serve as a catalyst. A final takeaway from this study is the need for acknowledging the purpose of strategic and interventionist language and message framing in public health messaging around heavily politicised medical treatments—attempts to keep public health messaging as scientific, objective, and non-persuasive as possible contributes itself to potential dismissal of those messages and therefore indirectly may foster the propagation of misinformation and conspiracism.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/covid4120132/s1>, Content analysis supplementary table all included.

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